Model answer to Q 1: New Calculations

Consider a parallel-plate transmission line (operating in the dominant TEM mode) with a width W = 1 mm and separation distance of h = 0.01 mm. Assume both plates are made from T = 0.01 mm thick copper, having a conductivity of 5.8 x 10^7 S/m. The line is embedded with a homogenous medium of PTFE, having a dielectric constant of 2.1, loss tangent of 2.1 x 10^{-4} and dielectric strength (breakdown field strength) of 10 kV/mm.

Unless otherwise stated, calculate the following at 2 GHz and include the units:

$$W = 1\text{ mm}, h = 0.01\text{ mm}, T = 0.01\text{ mm}, \sigma_0 = 5.8\text{ x }10^7\text{ S/m},$$
$$\varepsilon_r = 2.1, tan\delta = 2.1\text{x }10^{-4}, E_{break} = 10\text{ kV/mm}$$

i) Total DC resistance in Ω/m.

$$R[\Omega/\text{m}] = R_{TOP} + R_{BOTTOM} = 2R_{TOP} = \frac{2}{\sigma_0 T \cdot W}$$

$$R = 1.72\ \Omega/\text{m}$$

[2]

ii) Surface resistance and plate thickness to skin depth ratio.

$$R_s = \sqrt{\frac{\omega\mu}{2\sigma_o}} = 11.67\ \text{m}\Omega/\blacksquare \text{ and } \delta = \frac{1}{\sigma_0 R_s} = 1.48\ \mu\text{m and } \frac{T}{\delta} = 6.8$$

[4]

iii) Distributed resistance.

$$R[\Omega/\text{m}] = R_{TOP} + R_{BOTTOM} = 2R_{TOP} \sim \frac{2}{\sigma_0 \delta \cdot W} = \frac{2R_s}{W}$$

$$R = 23.34\ \Omega/\text{m}$$

[3]

iv) Distributed inductance.

$$L \approx \mu \frac{h}{W} = \mu_o \frac{h}{W} = 12.566\ \text{nH/m}$$

[2]

v) Distributed conductance.

$$\sigma_{eff} = \omega\varepsilon_o\varepsilon_r tan\delta = 49.07\ \mu\text{S/m}$$

$$G[S/m] \sim \sigma_{eff} \frac{W}{h} = 4.907\ \text{mS/m}$$

[4]

vi) Distributed capacitance.

$$C \sim \varepsilon_0 \varepsilon_r \frac{W}{h} = 1.859\ \text{nF/m}$$

[2]

vii) Lossless characteristic impedance.

Answers

$$Z_0 \xrightarrow{Lossless} \sqrt{\frac{L}{C}} \approx \frac{\eta_o}{\sqrt{\varepsilon_r}} \cdot \frac{h}{W} = 2.6\ \Omega$$

[2]

viii) Lossy characteristic impedance.

$$Zo = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = 2.607 - j0.191\ \Omega$$

[3]

ix) Lossless phase velocity.

$$v_p = \frac{1}{\sqrt{LC}} = \frac{c}{\sqrt{\varepsilon_r}} = 2.1 \text{ x } 10^8 \text{ m/s}$$

[2]

x) Lossless phase constant.

$$\beta = \omega\sqrt{LC} = 60.736 \text{ rad/m}$$

[2]

xi) Attenuation constant due to the conductors only.

$$\alpha_{conductor} = \frac{R}{2Z_0} = 4.49 \text{ Np/m}$$

[2]

xii) Attenuation constant due to the dielectric only.

$$\alpha_{dielectric} = \frac{Z_0 G}{2} = 0.0064 \text{ Np/m}$$

[2]

xiii) Total attenuation constant and insertion loss in dB/m.

$$\alpha_{total} = \alpha_{conductor} + \alpha_{dielectric} = 4.4934 \text{ Np/m}$$

$$IL = 8.686\ \alpha_{total} = 39 \text{ dB/m}$$

[4]

xiv) Q-factor associated with the conductors only.

$$Q_{conductor} = \frac{\omega L}{R} = 6.766$$

[2]

xv) Q-factor associated with the dielectric only.

$$Q_{dielectric} = \frac{\omega C}{G} = 4{,}760$$

[2]

xvi) Overall Q-factor.

$$Q = \left(\frac{1}{Q_{conductor}} + \frac{1}{Q_{dielectric}}\right)^{-1} = 6.756$$

[2]

xvii) Maximum peak voltage.

Answers

$$V_{peak.MAX} = E_{break} \cdot h = 100 \text{ V}$$

[2]

xviii) Maximum peak power under impedance matched conditions.

$$P_{peak.MAX} = \frac{\left|V_{peak.MAX}\right|^2_{RMS}}{Zo} = 3.846 \text{ kW}$$

[2]

xix) Lowest-order transverse resonant mode frequency.

$$f_T \approx \frac{v_p}{\lambda_T} = \frac{c/\sqrt{\varepsilon_r}}{2\,W} = 103.5 \text{ GHz}$$

[4]

xx) Maximum frequency for TEM mode of operation.

TEM operates from DC up to the first-order Transverse Magnetic and Transverse Electric mode cut-off frequencies:

$$f_c^{TM1} = f_c^{TE1} \approx \frac{v_p}{\lambda_c^{TE1}} = \frac{c/\sqrt{\varepsilon_r}}{2h} = 10.35 \text{ THz}$$

[3]

Model answer to Q 2(a): Discussions in Class

This question relates to the classical skin effect of normal metals.

State the main application(s) in the microwave industry where aluminium is used at room temperature and at liquid nitrogen temperatures. Briefly explain the reasons for your answers.

Aluminium is used in (silicon & SiGe) ICs, because it is relatively easy and cheap to deposit aluminium. Also, it has the highest conductivity of any metal at the 77 K boiling point of liquid nitrogen. As a result, low noise amplifiers used in radio astronomy and radar systems are cooled-down to 77 K to give superior detector/receiver noise figure performances.

[3]

Model answer to Q 2(b): Derivation

Show how the classical skin depth can be calculated from the propagation constant.

Inside a normal metal, propagation constant, $\gamma = \alpha(1+j)$ and attenuation constant $\alpha = \frac{1}{\delta_o}$ and so classical skin depth $\delta_o = \frac{1}{\gamma}(1+j)$.

[2]

Model answer to Q 2(c): Derivation

Show how the complex skin depth can be calculated from the propagation constant and express the real and imaginary parts in terms of the classical skin depth.

$$\text{Complex Skin Depth, } \tilde{\delta} = \delta' - j\delta'' = \frac{1}{\gamma} = \frac{\delta_o}{(1+j)} = \frac{\delta_o}{2}(1-j) \text{ and } \delta' = \delta'' = \frac{\delta_o}{2}$$

[2]

Model answer to Q 2(d): New Derivation

Derive the simple expressions for both the surface resistance and surface inductance, in terms of the classical skin depth.

$$Z_{S0} = \frac{j\omega\,\mu_o}{\gamma} \equiv R_{S0} + j\omega L_{S0} \quad \text{and } \gamma = (1+j)/\delta_o \text{ and } \delta_o = \sqrt{\frac{2}{\omega\mu_o\sigma_o}}$$

Answers

$$R_{S0} = Re\{Z_{S0}\} = Re\left\{\frac{j\omega\,\mu_o}{\gamma}\right\} = Re\left\{\frac{j\omega\,\mu_o\delta_o}{(1+j)}\right\} = \frac{1}{\sigma_o\delta_o}$$

$$L_{S0} = \frac{Im\{Z_{S0}\}}{\omega} = \frac{1}{\omega}Im\left\{\frac{j\omega\,\mu_o}{\gamma}\right\} = \frac{1}{\omega}Im\left\{\frac{j\omega\,\mu_o\delta_o}{2}(1-j)\right\} = \mu_o\frac{\delta_o}{2}$$

[4]

Model answer to Q 2(e): New Calculations

Aluminium has a bulk DC conductivity of $\sigma_o = 3.8 \times 10^7$ S/m at room temperature. For an operating frequency of 1 GHz, calculate the following:

i) Classical skin depth.

$$\delta_o = \sqrt{\frac{2}{\omega\mu_o\sigma_o}} = 2.58\ \mu m$$

[2]

ii) Wavelength inside the metal.

$$\lambda = 2\pi\delta_o = 16\ \mu m$$

[2]

iii) Surface impedance.

$$Z_{S0} = \frac{1}{\sigma_o\delta_o}(1+j) = 0.01(1+j)\Omega$$

[2]

iv) Surface resistance.

$$R_{S0} = Re\{Z_{S0}\} = 0.01\ \Omega$$

[1]

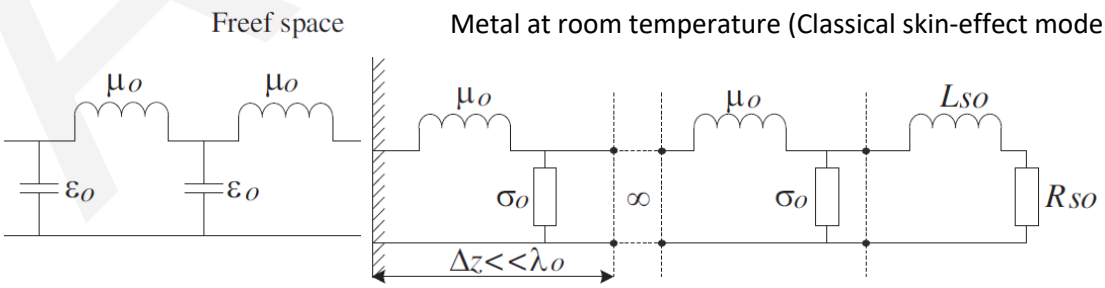
v) Surface inductance.

$$L_{S0} = \frac{Im\{Z_{S0}\}}{\omega} = 1.62\ pH$$

[2]

Model answer to Q 2(f): Discussions in Class

Draw the telegraphers equation's one-dimensional equivalent circuit models for propagation of a TEM wave both in free space at normal incidence to a metal boundary and also within the metal. Clearly label all the distributed lumped elements and give their units.



μ_o [H/m] ; ε_o [F/m] ; σ_o [S/m] ; R_{S0} [Ω] ; L_{S0}[H]

[5]

Answers

Model answer to Q3(a): New Derivations and Computed Example

An ideal air-filled rectangular waveguide has a guided-wavelength given by the following expression:

$$\lambda_g = \frac{\lambda_o}{\sqrt{1-\left(\frac{f_c}{f_o}\right)^2}} \qquad (5.1)$$

All variables have their usual meaning.

In addition, for the TE_{101} mode, the unloaded Q-factor for an air-filled rectangular waveguide resonant cavity is given by the following expression:

$$Q_u|_{TE101} \cong \frac{Volume}{Area \times (\delta_o/2)} \qquad (5.2)$$

All variables have their usual meaning.

a) For a half-height waveguide: height dimension b is half that of the width dimension a:

(i) Using (3.1), derive an expression for the length *l* of the cavity in terms of *a* and the various frequency terms.

$$for \quad half-height: b = \frac{a}{2}$$

$$\lambda_c = 2a \quad \therefore f_c = \frac{c}{2a} \quad \therefore a = \frac{\lambda_c}{2} = \frac{c}{2f_c} \quad also \quad \lambda_o = \frac{c}{f_o} \quad \therefore a = \frac{\lambda_o}{2}\left(\frac{f_o}{f_c}\right)$$

$$for \quad TE_{101}\ \mathrm{mode}: l \equiv \frac{\lambda g}{2} = \frac{\lambda_o/2}{\sqrt{1-\left(\frac{f_c}{f_o}\right)^2}} = \frac{a}{\left(\frac{f_o}{f_c}\right)\sqrt{1-\left(\frac{f_c}{f_o}\right)^2}} = \frac{a}{\sqrt{\left(\frac{f_o}{f_c}\right)^2-1}}$$

[5]

(ii) Using (i), derive an expression for the internal volume of the cavity.

$$Volume \equiv a\,b\,l = \frac{a^3}{2\sqrt{\left(\frac{f_o}{f_c}\right)^2-1}}$$

[3]

(iii) Using (i), derive an expression for the internal area of the cavity.

$$Area \equiv 2(a\,l + a\,b + b\,l) = 2a^2\left[\frac{1}{2} + \frac{1}{\sqrt{\left(\frac{f_o}{f_c}\right)^2-1}} + \frac{1}{2\sqrt{\left(\frac{f_o}{f_c}\right)^2-1}}\right] = a^2\left[1 + \frac{3}{\sqrt{\left(\frac{f_o}{f_c}\right)^2-1}}\right]$$

[4]

(iv) Using (3.2) and assuming that $f_o/f_c = \sqrt{2}$, derive an expression for the unloaded-Q-factor in terms of a and classical skin depth.

$$Volume = \frac{a^3}{2\sqrt{\left(\frac{f_o}{f_c}\right)^2 - 1}} \rightarrow \frac{a^3}{2} \quad and \quad Area = a^2\left[1 + \frac{3}{\sqrt{\left(\frac{f_o}{f_c}\right)^2 - 1}}\right] \rightarrow 4a^2$$

$$\therefore Q_u\big|_{TE101} \cong \frac{2\,Volume}{\delta_o\ Area} \rightarrow \frac{a}{4\delta_o}$$

[4]

(v) Using (iv), calculate the unloaded-Q-factor for a 15.5 GHz resonant cavity made with copper walls having a DC bulk conductivity of $\sigma_o = 5.8 \times 10^7$ S/m.

$$\lambda_o = \frac{c}{f_o} = 19.355mm \quad and \quad a = \frac{\lambda_o}{2}\left(\frac{f_o}{f_c}\right) = 13.686mm$$

$$\delta_o = \sqrt{\frac{2}{\omega_o \mu_o \sigma_o}} = \sqrt{\frac{2}{2\pi 15.5\times10^9 \times 4\pi \times 10^{-7} \times 5.8\times10^7}} = 0.531\mu m$$

$$\therefore Q_u\big|_{TE101} \cong \frac{a}{4\delta_o} = 6{,}446$$

[5]

Model answer to Q3(b): New Derivation and Computed Example

For a cubical cavity (i.e. all internal dimensions are equal) derive an expression for the unloaded-Q-factor in terms of a and classical skin depth and show that this has a 33.333% higher unloaded Q-factor.

$$Volume = a^3 \quad and \quad Area = 6a^2$$

$$\therefore Q_u\big|_{TE101} \cong \frac{2\,Volume}{\delta_o\ Area} \rightarrow \frac{a}{3\delta_o} \quad which\ is\ 4/3 = 1.3333333\ higher\ than\ with\ half-height$$

[4]

Model answer to Q4(a): Bookwork

Briefly describe the various steps needed to derive a transfer function using the Mason's non-touching loop rule and state the general form of the corresponding equation for a transfer function.

Mason's Non-Touching Loop Rule has the following steps:

First identify the input and output nodes you are interested in. For example, to calculate S11 of a network you need to concentrate on the nodes *a1* (input) and *b1* (output)

Second, identify all the PATHS going from the input to output nodes.

Third, identify any loops in the network, and note whether they touch any of your paths.

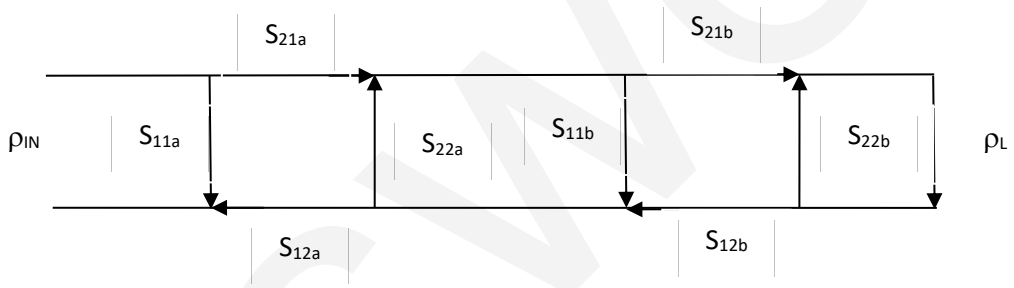
Fourth, apply the following equation:

$$Transfer\ Function = \frac{Output\ Variable}{Input\ Variable} = \frac{\sum_{n=1,2,3}^{\infty} P_n \cdot \begin{pmatrix} 1 - \Sigma 1st\ order\ loops\ not\ touching\ P_n \\ + \Sigma 2nd\ order\ loops\ not\ touching\ P_n \\ - \Sigma 3rd\ order\ loops\ not\ touching\ P_n \\ + \ldots \end{pmatrix}}{\begin{matrix} 1 - \Sigma 1st\ order\ loops \\ + \Sigma 2nd\ order\ loops \\ - \Sigma 3rd\ order\ loops \\ + \ldots \end{matrix}}$$

[5]

Model answer to Q4(b): New Derivation

Two amplifiers are connected in cascade, with the final output stage terminated with a load impedance having an arbitrary value. The amplifiers can be defined by the scattering (S)-parameter matrices $[S_a]$ and $[S_b]$, and the load impedance can be expressed by its voltage-wave reflection coefficient ρ_L. Draw the corresponding signal flow graph and determine the exact expression for the overall input voltage-wave reflection coefficient ρ_{IN}.



Paths (P) from nodes a1 to b1

P1 = S11a

P2 = S21a S21b ρ_L S12b S12a

P3 = S21a S11b S12a

First Order Loops (L)

FL1 = S22b ρ_L not touching P1 or P3

FL2 = S21b ρ_L S12b S22a not touching P1

FL3 = S22a S11b not touching P1

Second Order loop

SL = FL1 FL3

$$\rho_{IN} = b1/a1 = \frac{P1[1 - (FL1 + FL2 + FL3) + SL] + P2[1-0] + P3[1 - FL1]}{1 - (FL1 + FL2 + FL3) + SL}$$

[14]

Model answer to Q4(c): Computed Example

For simplicity, phase angle is ignored and the amplifiers in 4(b) are assumed to be identical. Given the following:

$$[S_a] = [S_b] = \begin{bmatrix} 0.1 & 0.1 \\ 10 & 0.1 \end{bmatrix} \quad \text{and} \quad \rho_L = 0.5$$

Calculate the overall input return loss for the circuit in 4(b) and comment on how this has changed by introducing the amplifiers.

ρ_{IN} = 0.68405/0.8905 = 0.76816 and overall Input Return Loss is 2.291 dB. The introduction of the amplifiers has degraded the overall Input Return Loss from the value of Load Return Loss = 6 dB.

[6]

Answers